

MASSENA, NY, USA

WMO: 726223

Lat: 44.936N

Lon: 74.846W

Elev: 214

StdP: 14.58

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94725

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.4	-7.6	-21.0	1.7	-11.5	-15.5	2.4	-6.2	26.9	28.7	24.4	25.9	4.9	220	0.571

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.4	87.5	72.6	84.6	70.9	82.1	69.5	75.2	83.8	73.3	81.3	71.6	79.2	10.9	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.3	120.6	80.3	70.5	113.2	78.2	68.9	107.0	76.8	38.7	83.9	37.0	81.7	35.5	79.3	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.3	19.6	17.8	DB	-20.8	91.8	7.0	2.3	-25.8	93.4	-29.9	94.8	-33.9	96.1	-39.0	97.7	(4)
			WB	-21.1	78.8	6.9	2.3	-26.1	80.4	-30.1	81.8	-34.0	83.1	-39.0	84.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	44.8	16.1	18.8	29.0	43.5	56.4	65.3	69.9	68.0	60.5	48.3	36.3	24.3	(6)
		DBStd	20.95	13.88	12.05	11.25	8.74	7.77	6.83	5.32	5.47	7.44	8.32	9.31	11.70	(7)
		HDD50	4200	1052	875	656	234	25	0	0	6	136	419	797		(8)
		HDD65	7807	1517	1294	1117	648	286	79	15	33	175	520	861	1262	(9)
		CDD50	2307	0	0	4	37	223	459	616	557	320	82	8	1	(10)
		CDD65	441	0	0	0	2	19	88	166	125	39	2	0	0	(11)
		CDH74	4022	0	0	5	28	265	834	1466	1027	372	25	0	0	(12)
		CDH80	1052	0	0	0	4	62	225	415	258	88	0	0	0	(13)
(14)	Wind	WSAvg	7.7	8.7	8.8	8.7	9.1	7.9	6.5	6.4	5.8	6.1	7.5	8.1	8.3	(14)
(15)	Precipitation	PrecAvg	34.5	2.3	2.0	2.3	2.7	2.8	3.1	3.3	3.4	3.5	3.0	3.1	3.0	(15)
(16)		PrecMax	47.4	5.4	5.6	4.6	7.4	6.4	6.3	7.2	7.7	7.2	6.8	6.1	6.7	(16)
(17)		PrecMin	24.2	0.2	0.3	0.7	0.4	1.1	0.6	1.2	0.8	0.9	0.4	1.1	0.9	(17)
(18)		PrecStd	4.6	1.2	1.1	1.0	1.5	1.3	1.4	1.5	1.3	1.5	1.3	1.0	1.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	54.9	52.2	67.5	78.0	86.2	89.5	90.9	89.9	87.8	77.5	66.5	56.9		(19)
(20)		0.4% MCWB	50.9	44.7	55.1	60.9	69.4	74.1	75.8	73.2	72.4	63.3	57.0	50.2		(20)
(21)		2% DB	45.4	43.9	56.1	70.3	80.9	85.6	87.3	85.8	82.1	71.9	61.2	48.9		(21)
(22)		2% MCWB	42.3	39.5	47.6	55.3	65.1	71.1	72.9	71.7	70.0	61.9	53.8	45.2		(22)
(23)		5% DB	39.1	38.9	49.7	64.4	76.2	82.2	84.4	82.6	78.3	66.6	55.5	43.5		(23)
(24)		5% MCWB	36.2	35.8	42.7	51.7	61.8	68.9	71.4	70.0	67.5	58.2	49.2	40.2		(24)
(25)		10% DB	34.9	35.9	44.2	59.1	71.9	79.1	81.9	80.0	74.4	62.5	50.5	38.9		(25)
(26)		10% MCWB	32.8	33.1	39.0	49.0	59.6	66.8	70.0	68.9	65.4	55.8	45.5	36.0		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	50.5	46.2	58.3	63.4	71.4	76.5	78.3	76.4	75.1	67.7	59.2	51.9		(27)
(28)		0.4% MCDB	53.9	50.2	66.1	74.3	83.2	86.9	87.8	85.4	83.6	74.5	63.7	55.6		(28)
(29)		2% WB	42.3	39.9	49.1	57.8	67.9	73.5	75.7	74.1	71.6	62.9	55.2	45.6		(29)
(30)		2% MCDB	45.1	43.2	55.1	66.6	77.8	82.5	83.8	82.2	79.6	69.2	60.9	48.6		(30)
(31)		5% WB	36.5	36.2	43.7	53.5	64.7	71.3	73.5	72.2	69.1	59.3	50.3	40.7		(31)
(32)		5% MCDB	39.0	38.8	48.8	61.5	72.5	79.2	81.2	79.5	76.0	65.9	55.0	43.6		(32)
(33)		10% WB	32.9	33.3	39.3	49.9	61.7	69.0	71.8	70.3	66.8	56.3	46.0	36.4		(33)
(34)		10% MCDB	34.7	35.3	43.7	58.5	69.3	76.2	79.3	77.1	73.0	61.6	50.1	38.6		(34)
(35)	Mean Daily Temperature Range	MDBR	17.7	18.6	17.9	19.7	21.6	20.5	20.4	20.7	21.4	18.1	15.8	14.8		(35)
(36)		5% DB MCDBR	20.5	19.2	25.5	29.8	28.7	24.9	23.0	24.0	25.5	25.8	24.9	18.2		(36)
(37)		5% DB MCWBR	18.7	16.6	18.0	17.8	15.4	13.1	11.9	12.1	14.5	16.5	18.8	15.8		(37)
(38)		5% WB MCDBR	20.2	19.0	24.1	25.3	24.2	20.9	20.2	20.1	22.3	23.0	22.8	17.8		(38)
(39)		5% WB MCWBR	18.6	16.9	18.2	17.7	15.0	12.4	11.7	11.5	13.6	16.1	18.4	16.0		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.270	0.294	0.329	0.354	0.387	0.422	0.434	0.412	0.370	0.343	0.317	0.286		(40)
(41)		taud	2.397	2.333	2.375	2.395	2.344	2.257	2.226	2.287	2.396	2.471	2.473	2.403		(41)
(42)		Ebn at Noon	271	283	287	288	280	269	264	266	269	262	251	251		(42)
(43)		Edh at Noon	23	30	33	36	39	42	43	39	32	26	22	21		(43)
(44)	All-Sky Solar Radiation	RadAvg	436	727	1125	1391	1630	1762	1800	1595	1264	740	460	321		(44)
(45)		RadStd	52	75	99	162	171	140	154	75	107	73	52	31		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page